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Oiowe chady

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Templatki (1)

.bashrc 11 lines

```
c() {
    g++ -std=c++20 -Wall -Wno-sign-conversion -Wextra -Wshadow -
        Wconversion -Wfloat-equal \
    -fsanitize=address,undefined -DLOCAL -DDEBUG -D_GLIBCXX_DEBUG
        -ggdb3 $1.cpp -o $1
}

nc() {
    g++ -std=c++20 -DLOCAL -O3 -static $1.cpp -o $1
}
```

```
alias cp='cp -i'
alias mv='mv -i'
```

.vimrc 7 lines

```
set nocompatible
filetype indent on
set autoindent
set number
syntax on
set ts=4 sw=4
set rnu
```

main.cpp 24 lines

```
#include <bits/stdc++.h>
using namespace std;
#ifdef DEBUG
auto&operator<<((auto &o, pair<auto, auto> p) {o << "(" << p.
    first << ", " << p.second << ")"; return o;}
auto operator<<((auto&o,auto x)->decltype(x.end()),o){o<<"{"; for
    (auto e : x) o<<e<<" "; return o<<"}";}
#define debug(X) cerr << "["#X"]": " << X << '\n';
#else
#define cerr if(0)cout
#define debug(X) ;
#endif
using ll = long long;
#define all(v) (v).begin(), (v).end()
#define ssize(x) int(x.size())
#define fi first
#define se second
```

```
1 #define mp make_pair
1 #define eb emplace_back
1
1 int main() {
2     ios_base::sync_with_stdio(false); cin.tie(nullptr);
2
2     return 0;
6 }
```

gen.cpp
Opis: Dodatek do generowania testów, lub innego losowania. Ziarno można zmieniać np. rng.seed(atoi(argv[1]));

```
mt19937 rng(random_device{}());
10 int rd(int l, int r) {
10     return uniform_int_distribution<int>(l, r)(rng);
10 }
```

10 **Wzorki (2)**

2.1 Fibonacci

$f_0 = 0, f_1 = 1, f_n = f_{n-1} + f_{n-2}$, dla $n \geq 2$.

$$f_n = \frac{\alpha^n - \beta^n}{\sqrt{5}} = \frac{\alpha^n - \beta^n}{\alpha - \beta},$$

dla $\alpha = \frac{1+\sqrt{5}}{2}, \beta = \frac{1-\sqrt{5}}{2}$. Jako, że $|\beta| < 1$ to

$$f_n \approx \frac{1}{\sqrt{5}} \alpha^n.$$

Niech $F(x) = \sum_{n=0}^{\infty} f_n x^n$, wtedy

$$F(x) = \frac{x}{1 - x - x^2}.$$

$$f_1 + f_2 + \dots + f_n = f_{n+2} - 1$$

$$\text{NWD}(f_n, f_m) = f_{\text{NWD}(n, m)}.$$

2.2 Liczby Catalana

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \dots + C_{n-2} C_1 + C_{n-1} C_0$$

$$C_{n+1} = \frac{2(2n+1)}{n+2} C_n$$

2.3 Funkcje tworzące

$$\frac{1}{1 - ax} = \sum a^n x^n$$

$$F(x) = \frac{x}{1 - x - x^2}$$

$$C(x) = \frac{1 - \sqrt{1 - 4x}}{2x}$$

$$\exp(x) = \sum_{n \geq 0} \frac{x^n}{n!}$$

$$-\ln(1 - x) = \sum_{n \geq 1} \frac{x^n}{n}$$

$$\frac{1}{(1 - x)^k} = \sum_{n \geq 0} \binom{n + k - 1}{n} x^n$$

$$\prod_{n \geq 1} (1 - x^n) = 1 + \sum_{k \geq 1} (-1)^k \left(x^{\frac{k(3k+1)}{2}} + x^{\frac{k(3k-1)}{2}} \right)$$

2.4 Geometria

2.4.1 Obrót płaszczyzny wokół punktu (0,0).

$$R_\theta \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

W szczególności

$$R_{\frac{\pi}{4}} \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} x + y \\ x - y \end{bmatrix}.$$

2.5 Divide and Conquer DP

Jeżeli

$$dp(i, j) = \min_{0 \leq k \leq j} dp(i - 1, k - 1) + C(k, j),$$

oraz

$$C(a, c) + C(b, d) \leq C(a, d) + C(b, c),$$

dla $a \leq b \leq c \leq d$, to $opt(i, j) \leq opt(i, j + 1)$, gdzie $opt(i, j)$ to indeks k minimalizujący powyższy wzór. Można w takim razie użyć divide and conquer do przyspieszenia.

2.6 Zasada włączeń i wyłączeń

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{i_1, i_2, \dots, i_k} (-1)^k |A_{i_1} \cap A_{i_2} \dots A_{i_k}|$$

2.7 Funkcje arytmetyczne

$$\varphi(ab) = \varphi(a) \cdot \varphi(b) \cdot \frac{d}{\varphi(d)},$$

gdzie $d = \text{NWD}(a, b)$. Jeżeli $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$, to

$$\varphi(n) = n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right).$$

$$\varphi(n) = \sum_{d|n} \varphi(d)$$

Niech $\phi(n) = \sum_{i=1}^n \varphi(i)$, wtedy

$$\phi(n) = \frac{n(n+1)}{2} - \sum_{i=2}^n \phi\left(\left\lfloor \frac{n}{i} \right\rfloor\right)$$

Matma (3)

ext-euclid.cpp

Opis: $O(\log n)$. Zwraca trójkę $[d, x, y]$, że $d = \text{NWD}(a, b)$ i $ax + by = d$.

```
tuple<ll, ll, ll> euclid(ll a, ll b) {
    if (b == 0) return make_tuple(a, 1, 0);
    auto [d, x, y] = euclid(b, a%b);
    return make_tuple(d, y, x - (a/b) * y);
}
```

phi.cpp

Opis: Liczy $\phi(n)$ w $O(\sqrt{n})$.

```
int phi(int n) {
    int result = n;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0)
                n /= i;
            result -= result / i;
        }
    }
    if (n > 1)
        result -= result / n;
    return result;
}
```

phi-to-n.cpp

Opis: Liczy $\phi(a)$ dla każdego $a \leq n$ w $O(n \log \log n)$.

```
void phi_1_to_n(int n) {
    vector<int> phi(n + 1);
    for (int i = 0; i <= n; i++)
        phi[i] = i;

    for (int i = 2; i <= n; i++) {
        if (phi[i] == i) {
            for (int j = i; j <= n; j += i)
                phi[j] -= phi[j] / i;
        }
    }
}
```

sum-phi.cpp

Opis: $O(n^{2/3})$ dla $HI = n^{2/3}$. Liczy $\sum_{i=1}^n \varphi(n)$.

```
struct sieve {
    vector<int> phi;
    sieve(int n) {
        phi.resize(n+1);
        for (int i = 0; i <= n; ++i) phi[i] = i;

        for (int i = 2; i <= n; ++i) {
            if (phi[i] == i) {
                for (int j = i; j <= n; j += i)
                    phi[j] -= phi[j]/i;
            }
        }
    }
};

const int HI = 1'000'003;
sieve S(HI);

unordered_map<int, int> mem;
vector<int> small(HI);

int sum_phi(int n) {
    if (n < HI) return small[n];
    if (mem[n] > 0) return mem[n];

    int res = 1l(n) * (n+1) / 2 % mod;
    for (int l = 2, r; l <= n; l = r) {
        int k = n / l;
        r = n / k + 1;
        res = sub(res, mul((r - 1), sum_phi(k)));
    }
    return mem[n] = res;
}
```

crt.cpp

Opis: $O(\log n)$. Zwraca takie x , że $x \equiv a \pmod m$ i $x \equiv b \pmod n$. Mogą nie być względnie pierwsze ale wtedy assert wywali.

```
ll crt(ll a, ll m, ll b, ll n) {
    if(n > m) swap(a, b), swap(m, n);
    auto [d, x, y] = euclid(m, n);
    assert((a - b) % d == 0);
    ll ret = (b - a) % n * x % n / d * m + a;
    return ret < 0 ? ret + m * n / d : ret;
}
```

xor-base.cpp

Opis: $O(nB)$ gdzie B to liczba bitów. Zwraca minimalny zbiór b taki że każdy element z x można zapisać jako xor pewnych elementów b .

```
vector<ll> xor_base(const vector<ll> &x) {
    vector<ll> basis;
    for (ll a: x) {
        ll A = a;
        for (ll b: basis) A = min(A, A^b);
        if (A) basis.push_back(A);
    }
    return basis;
}
```

fft.cpp

Opis: Mnożenie wielomianów w $O(n \log n)$.

```
using ld = long double;
using cd = complex<long double>;
const double PI = acos(-1);
```

```
void fft(vector<cd> &a, int n, bool invert = false) {
    assert((n&(n-1)) == 0);
    a.resize(n);
    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n/2;
        for (; bit&j; bit /= 2)
            j ^= bit;
        j ^= bit;

        if (i < j) swap(a[i], a[j]);
    }

    for (int len = 2; len <= n; len *= 2) {
        ld ang = 2 * PI / len * (invert ? -1 : 1);
        cd r_len(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            cd r(1);
            for (int j = 0; j < len/2; ++j, r *= r_len) {
                cd u = a[i + j], v = r * a[i + j + len/2];
                a[i+j] = u + v;
                a[i+j+len/2] = u - v;
            }
        }
    }

    if (invert)
        for (cd &x : a) x /= n;
}
```

```
vector<ld> conv(vector<ld> a, vector<ld> b) {
    if (a.empty() || b.empty()) return {};
```

```
    vector<cd> fa(all(a)), fb(all(b));
    int n = ssize(a) + ssize(b) - 1;
    int sz = 1;
    while (sz < ssize(a) + ssize(b) - 1) sz *= 2;
    fft(fa, sz);
    fft(fb, sz);
    for (int i = 0; i < sz; ++i) fa[i] *= fb[i];
    fft(fa, sz, true);
```

```
    vector<ld> result(n);
    for (int i = 0; i < n; ++i)
        result[i] = round(fa[i].real());
    return result;
}
```

mod.cpp

Opis: operacje mod 998244353.

```
const int mod = 998244353;
```

```
int add(int a, int b) {
    a += b;
    return a >= mod ? a-mod : a;
}
```

```
int sub(int a, int b) {
    return add(a, mod-b);
}
```

```
int mul(int a, int b) {
    return int(((ll)a*b)%mod);
}
```

```
int fpow(int a, int b) {
    int res = 1;
    while (b) {
```

```
    if (b&1) {
        res = mul(a, res);
        b--;
    }
    a = mul(a, a);
    b /= 2;
}

return res;
}

int inv(int x) {
    return fpow(x, mod-2);
}
```

ntt.cpp

Opis: $O(n \log n)$. Mnożenie wielomianów mod 998244353.

```
#include "mod.cpp" //keep-include

const int g = 3;
const int g_1 = 332748118;

using vi = vector<int>;

void ntt(vi &a, int n, bool invert = false) {
    assert((n&(n-1)) == 0);
    a.resize(n);
    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n/2;
        for ( ; bit&j; bit /= 2)
            j ^= bit;
        j ^= bit;

        if (i < j) swap(a[i], a[j]);
    }

    for (int len = 2; len <= n; len *= 2) {
        int r_len = fpow(invert ? g_1 : g, (mod-1) / len);
        for (int i = 0; i < n; i += len) {
            int r = 1;
            for (int j = 0; j < len/2; ++j, r = mul(r, r_len)) {
                int u = a[i + j], v = mul(r, a[i + j + len/2]);
                a[i+j] = add(u, v);
                a[i+j+len/2] = sub(u, v);
            }
        }
    }

    if (invert) {
        int inv_n = inv(n);
        for (int &x : a) x = mul(x, inv_n);
    }
}

vi conv(vi a, vi b) {
    if (a.empty() || b.empty()) return {};
    int n = ssize(a) + ssize(b) - 1;
    int sz = 1;
    while (sz < ssize(a) + ssize(b) - 1) sz *= 2;
    ntt(a, sz);
    ntt(b, sz);
    for (int i = 0; i < sz; ++i) a[i] = mul(a[i], b[i]);
    ntt(a, sz, true);
    a.resize(n);
    return a;
}
```

poly.cpp

Opis: Operacje na wielmianach mod 998244353. Można przepisać co się chce, przy funkcjach jest napisane co potrzeba. deriv, integr w $O(n)$. inv, log, exp w $O(n \log n)$.

```
#include "ntt.cpp" //keep-include

using vi = vector<int>;
vi mod_xn(const vi &v, int n) { //ZAWSZE
    return vi(v.begin(), v.begin() + min(n, ssize(v)));
}

vi add(vi a, const vi &b) {
    a.resize(max(ssize(a), ssize(b)));
    for (int i = 0; i < ssize(b); ++i)
        a[i] = add(a[i], b[i]);
    return a;
}

vi sub(vi a, const vi &b) { //ZAWSZE
    a.resize(max(ssize(a), ssize(b)));
    for (int i = 0; i < ssize(b); ++i)
        a[i] = sub(a[i], b[i]);
    return a;
}

vi deriv(vi p) {
    for (int i = 1; i < ssize(p); ++i) p[i] = mul(p[i], i);
    if (ssize(p)) p.erase(p.begin());
    return p;
}

vi integr(vi p) {
    for (int i = 0; i < ssize(p); ++i)
        p[i] = mul(p[i], inv(i+1));
    p.insert(p.begin(), 0);
    return p;
}

vi inv(vi p, int n) {
    assert(ssize(p) && p[0] != 0);
    if (n == 1)
        return {inv(p[0])};
    p.resize(n);
    vi _p = p;
    for (int i = 1; i < n; i += 2) _p[i] = sub(0, p[i]);

    vi t = mod_xn(conv(p, _p), n);
    for (int i = 1; i < n; i += 2) assert(t[i] == 0);
    for (int i = 0; i < n; i += 2)
        t[i/2] = t[i];
    t = inv(t, (n+1)/2);
    t.resize(n);
    for (int i = (n-1)/2; i >= 0; --i) {
        t[2*i] = t[i];
        if(i) t[i] = 0;
    }
    return mod_xn(conv(_p, t), n);
}

vi log(const vi &p, int n) { //DERIV, INV, INTEGR
    assert(ssize(p) && p[0] == 1);
    return integr(mod_xn(conv(deriv(mod_xn(p, n)), inv(p, n)), n
        -1));
}

vi exp(const vi &p, int n) { //ADD, LOG
    assert(p.empty() || p[0] == 0);
    vi q = {1};
```

```
    for (int x = 1; x < n; x *= 2)
        q = mod_xn(conv(q, add(sub({1}, log(q, 2*x)), mod_xn(p, 2*x
            )), 2*x);
    q.resize(n);
    return q;
}

vi powi(vi p, int k, int n) { //LOG, EXP
    p = mod_xn(p, n);
    int cnt = 0;
    while(cnt < ssize(p) and !p[cnt])
        ++cnt;
    if(ll(cnt) * k >= n)
        return {};
    p.erase(p.begin(), p.begin() + cnt);
    if(p.empty())
        return k ? vi{} : vi{1};
    int powi0 = fpow(p[0], k);
    int inv0 = inv(p[0]);
    for (int &x : p) x = mul(x, inv0);
    p = log(p, n - cnt * k);
    for (int &x : p) x = mul(x, k);
    p = exp(p, n - cnt * k);
    for (int &x : p) x = mul(x, powi0);
    vi t(cnt * k, 0);
    p.insert(p.begin(), t.begin(), t.end());
    return p;
}

vi sqrt(const vi &p, int n) { //POW
    assert(ssize(p) && p[0] == 1);
    return powi(p, inv(2), n);
}
```

poly2d.cpp

Opis: Operacje na wielmianach dwóch zmiennych mod 998244353. Mnożenie w $O(mn \log(m+n))$. Dzielenie zwraca vector współczynników przy x^{n-1} , w czasie $O(mn \log^2(n+m))$.

```
#include "poly.cpp" //keep-include

void ntt(vector<vi> &a, int n, int m, bool inverse = false) {
    a.resize(n);
    for (int i = 0; i < ssize(a); ++i)
        ntt(a[i], m, inverse);
    for (int j = 0; j < ssize(a[0]); ++j) {
        vi c;
        for (int i = 0; i < ssize(a); ++i)
            c.eb(a[i][j]);
        ntt(c, n, inverse);
        for (int i = 0; i < ssize(a); ++i)
            a[i][j] = c[i];
    }
}

vector<vi> conv(vector<vi> a, vector<vi> b, int n, int m) {
    if (a.empty() || b.empty()) return {};
    int l1 = ssize(a) + ssize(b) - 1, sz1 = 1 << __lg(2*l1-1);
    int l2 = ssize(a[0]) + ssize(b[0]) - 1, sz2 = 1 << __lg(2*l2
        -1);

    ntt(a, sz1, sz2);
    ntt(b, sz1, sz2);
    for (int i = 0; i < sz1; ++i)
        for (int j = 0; j < sz2; ++j) a[i][j] = mul(a[i][j], b[i][j
            ]);
    ntt(a, sz1, sz2, true);
    a.resize(n);
    for (int i = 0; i < n; ++i) a[i].resize(m);
```

```
        return a;
    }

vi div(vector<vi> b, vector<vi> a, int n, int m) {
    if (n == 1) {
        return conv(b[0], inv(a[0], m));
    }
    vector<vi> q = a;
    for (int i = 1; i < n; i += 2)
        for (int j = 0; j < ssize(a[i]); ++j) q[i][j] = sub(0, q[i-1][j]);

    vector<vi> v = conv(a, q, n, 2*m);
    debug(v);
    q = conv(b, q, n, 2*m);
    for (int i = 0; i < ssize(v); i += 2) {
        for (int j = 0; j < ssize(v[i]); ++j) {
            v[i/2][j] = v[i][j];
        }
    }
    v.resize((n+1)/2);

    if (n%2 == 0) {
        for (int i = 1; i < n; i += 2)
            for (int j = 0; j < 2*m; ++j)
                q[i/2][j] = q[i][j];
        q.resize(n/2);
        return div(q, v, n/2, 2*m);
    }
    else {
        for (int i = 0; i < n; i += 2)
            for (int j = 0; j < 2*m; ++j)
                q[i/2][j] = q[i][j];
        q.resize((n+1)/2);
        return div(q, v, (n+1)/2, 2*m);
    }
}
```

lis.cpp

Opis: $O(n \log n)$. Zwraca długość najdłuższego podciągu rosnącego.

```
int LIS(vector<int> A)
{
    vector<int> vec;
    for(auto& v : A)
        if(vec.size() == 0 || v > vec.back())
            vec.push_back(v);
        else
            *lower_bound(vec.begin(), vec.end(), v) = v;
    return vec.size();
}
```

mo.cpp

Opis: $O(n \log n)$. Sort do MO. Sortuje przedziały $[a.x, a.y]$.

```
const int k = 500;
sort(events.begin(), events.end(), [&](type a, type b){
    if(a.x/k < b.x/k) return true;
    if(a.x/k > b.x/k) return false;
    if((a.x/k)%2 == 1)
        return a.y < b.y;
    else
        return a.y > b.y;
});
```

sos.cpp

Opis: $O(l \cdot 2^l)$. Zwraca $sos[mask]$ równe sumie $s[m]$ po podzbiorach $mask$.

```
vector<int> SOS(int l, vector<int>& s)
```

```
{
    vector<int> sos(1<<l);
    for(int mask = 0; mask<(1<<l); mask++)
        sos[mask] = s[mask];
    for(int i=0; i<l; i++)
        for(int mask=0; mask<(1<<l); mask++)
            if((mask & (1<<i)) != 0)
                sos[mask] += sos[mask^(1<<i)];

    return sos;
}
```

big-num.cpp

Opis: bignumy.

```
typedef long long ll;
const int maxn = 1e2 + 14, lg = 15;

const int base = 10000000000;
const int base_digits = 9;
struct bigint {
    vector<int> a;
    int sign;
    /*<arpa>*/
    int size(){
        if(a.empty())return 0;
        int ans=(a.size()-1)*base_digits;
        int ca=a.back();
        while(ca)
            ans++,ca/=10;
        return ans;
    }
    bigint operator ^(const bigint &v){
        bigint ans=1,a=*this,b=v;
        while(!b.isZero()){
            if(b%2)
                ans*=a;
            a*=a,b/=2;
        }
        return ans;
    }
    string to_string(){
        stringstream ss;
        ss << *this;
        string s;
        ss >> s;
        return s;
    }
    int sumof(){
        string s = to_string();
        int ans = 0;
        for(auto c : s)  ans += c - '0';
        return ans;
    }
    /*</arpa>*/
    bigint() :
        sign(1) {
    }

    bigint(long long v) {
        *this = v;
    }

    bigint(const string &s) {
        read(s);
    }

    void operator=(const bigint &v) {
        sign = v.sign;
```

```
        a = v.a;
    }

    void operator=(long long v) {
        sign = 1;
        a.clear();
        if (v < 0)
            sign = -1, v = -v;
        for (; v > 0; v = v / base)
            a.push_back(v % base);
    }

    bigint operator+(const bigint &v) const {
        if (sign == v.sign) {
            bigint res = v;

            for (int i = 0, carry = 0; i < (int) max(a.size(), v.a.size()) || carry; ++i) {
                if (i == (int) res.a.size())
                    res.a.push_back(0);
                res.a[i] += carry + (i < (int) a.size() ? a[i] : 0);
                carry = res.a[i] >= base;
                if (carry)
                    res.a[i] -= base;
            }
            return res;
        }
        return *this - (-v);
    }

    bigint operator-(const bigint &v) const {
        if (sign == v.sign) {
            if (abs() >= v.abs()) {
                bigint res = *this;
                for (int i = 0, carry = 0; i < (int) v.a.size() || carry; ++i) {
                    res.a[i] -= carry + (i < (int) v.a.size() ? v.a[i] : 0);
                    carry = res.a[i] < 0;
                    if (carry)
                        res.a[i] += base;
                }
                res.trim();
                return res;
            }
            return -(v - *this);
        }
        return *this + (-v);
    }

    void operator*=(int v) {
        if (v < 0)
            sign = -sign, v = -v;
        for (int i = 0, carry = 0; i < (int) a.size() || carry; ++i) {
            if (i == (int) a.size())
                a.push_back(0);
            long long cur = a[i] * (long long) v + carry;
            carry = (int) (cur / base);
            a[i] = (int) (cur % base);
            /*asm("divl %%ecx" : "=a"(carry), "=d"(a[i]) : "A"(cur), "c"(base));*/
        }
        trim();
    }

    bigint operator*(int v) const {
        bigint res = *this;
        res *= v;
```

```

    return res;
}

void operator+=(long long v) {
    if (v < 0)
        sign = -sign, v = -v;
    if (v > base) {
        *this = *this * (v / base) * base + *this * (v % base);
        return;
    }
    for (int i = 0, carry = 0; i < (int) a.size() || carry; ++i)
    {
        if (i == (int) a.size())
            a.push_back(0);
        long long cur = a[i] * (long long) v + carry;
        carry = (int) (cur / base);
        a[i] = (int) (cur % base);
        //asm("divl %%ecx" : "=a"(carry), "=d"(a[i]) : "A"(cur),
        "c"(base));
    }
    trim();
}

bigint operator*(long long v) const {
    bigint res = *this;
    res *= v;
    return res;
}

friend pair<bigint, bigint> divmod(const bigint &a1, const
    bigint &b1) {
    int norm = base / (b1.a.back() + 1);
    bigint a = a1.abs() * norm;
    bigint b = b1.abs() * norm;
    bigint q, r;
    q.a.resize(a.a.size());

    for (int i = a.a.size() - 1; i >= 0; i--) {
        r *= base;
        r += a.a[i];
        int s1 = r.a.size() <= b.a.size() ? 0 : r.a[b.a.size()];
        int s2 = r.a.size() <= b.a.size() - 1 ? 0 : r.a[b.a.size()
            - 1];
        int d = ((long long) base * s1 + s2) / b.a.back();
        r -= b * d;
        while (r < 0)
            r += b, --d;
        q.a[i] = d;
    }

    q.sign = a1.sign * b1.sign;
    r.sign = a1.sign;
    q.trim();
    r.trim();
    return make_pair(q, r / norm);
}

bigint operator/(const bigint &v) const {
    return divmod(*this, v).first;
}

bigint operator%(const bigint &v) const {
    return divmod(*this, v).second;
}

void operator/=(int v) {
    if (v < 0)
        sign = -sign, v = -v;
    for (int i = (int) a.size() - 1, rem = 0; i >= 0; --i) {

```

```

        long long cur = a[i] + rem * (long long) base;
        a[i] = (int) (cur / v);
        rem = (int) (cur % v);
    }
    trim();
}

bigint operator/(int v) const {
    bigint res = *this;
    res /= v;
    return res;
}

int operator%(int v) const {
    if (v < 0)
        v = -v;
    int m = 0;
    for (int i = a.size() - 1; i >= 0; --i)
        m = (a[i] + m * (long long) base) % v;
    return m * sign;
}

void operator+=(const bigint &v) {
    *this = *this + v;
}

void operator-=(const bigint &v) {
    *this = *this - v;
}

void operator*=(const bigint &v) {
    *this = *this * v;
}

void operator/=(const bigint &v) {
    *this = *this / v;
}

bool operator<(const bigint &v) const {
    if (sign != v.sign)
        return sign < v.sign;
    if (a.size() != v.a.size())
        return a.size() * sign < v.a.size() * v.sign;
    for (int i = a.size() - 1; i >= 0; i--)
        if (a[i] != v.a[i])
            return a[i] * sign < v.a[i] * sign;
    return false;
}

bool operator>(const bigint &v) const {
    return v < *this;
}

bool operator<=(const bigint &v) const {
    return !(v < *this);
}

bool operator>=(const bigint &v) const {
    return !(*this < v);
}

bool operator==(const bigint &v) const {
    return !(*this < v) && !(v < *this);
}

bool operator!=(const bigint &v) const {
    return *this < v || v < *this;
}

void trim() {
    while (!a.empty() && !a.back())
        a.pop_back();
    if (a.empty())
        sign = 1;
}

```

```

bool isZero() const {
    return a.empty() || (a.size() == 1 && !a[0]);
}

bigint operator-() const {
    bigint res = *this;
    res.sign = -sign;
    return res;
}

bigint abs() const {
    bigint res = *this;
    res.sign *= res.sign;
    return res;
}

long long longValue() const {
    long long res = 0;
    for (int i = a.size() - 1; i >= 0; i--)
        res = res * base + a[i];
    return res * sign;
}

friend bigint gcd(const bigint &a, const bigint &b) {
    return b.isZero() ? a : gcd(b, a % b);
}

friend bigint lcm(const bigint &a, const bigint &b) {
    return a / gcd(a, b) * b;
}

void read(const string &s) {
    sign = 1;
    a.clear();
    int pos = 0;
    while (pos < (int) s.size() && (s[pos] == '-' || s[pos] ==
        '+')) {
        if (s[pos] == '-')
            sign = -sign;
        ++pos;
    }
    for (int i = s.size() - 1; i >= pos; i -= base_digits) {
        int x = 0;
        for (int j = max(pos, i - base_digits + 1); j <= i; j++)
            x = x * 10 + s[j] - '0';
        a.push_back(x);
    }
    trim();
}

friend istream& operator>>(istream &stream, bigint &v) {
    string s;
    stream >> s;
    v.read(s);
    return stream;
}

friend ostream& operator<<(ostream &stream, const bigint &v)
{
    if (v.sign == -1)
        stream << '-';
    stream << (v.a.empty() ? 0 : v.a.back());
    for (int i = (int) v.a.size() - 2; i >= 0; --i)
        stream << setw(base_digits) << setfill('0') << v.a[i];
    return stream;
}

static vector<int> convert_base(const vector<int> &a, int
    old_digits, int new_digits) {
    vector<long long> p(max(old_digits, new_digits) + 1);

```

```
p[0] = 1;
for (int i = 1; i < (int) p.size(); i++)
    p[i] = p[i - 1] * 10;
vector<int> res;
long long cur = 0;
int cur_digits = 0;
for (int i = 0; i < (int) a.size(); i++) {
    cur += a[i] * p[cur_digits];
    cur_digits += old_digits;
    while (cur_digits >= new_digits) {
        res.push_back((int) (cur % p[new_digits]));
        cur /= p[new_digits];
        cur_digits -= new_digits;
    }
}
res.push_back((int) cur);
while (!res.empty() && !res.back())
    res.pop_back();
return res;
}

typedef vector<long long> vll;

static vll karatsubaMultiply(const vll &a, const vll &b) {
    int n = a.size();
    vll res(n + n);
    if (n <= 32) {
        for (int i = 0; i < n; i++)
            for (int j = 0; j < n; j++)
                res[i + j] += a[i] * b[j];
        return res;
    }

    int k = n >> 1;
    vll a1(a.begin(), a.begin() + k);
    vll a2(a.begin() + k, a.end());
    vll b1(b.begin(), b.begin() + k);
    vll b2(b.begin() + k, b.end());

    vll a1b1 = karatsubaMultiply(a1, b1);
    vll a2b2 = karatsubaMultiply(a2, b2);

    for (int i = 0; i < k; i++)
        a2[i] += a1[i];
    for (int i = 0; i < k; i++)
        b2[i] += b1[i];

    vll r = karatsubaMultiply(a2, b2);
    for (int i = 0; i < (int) a1b1.size(); i++)
        r[i] -= a1b1[i];
    for (int i = 0; i < (int) a2b2.size(); i++)
        r[i] -= a2b2[i];

    for (int i = 0; i < (int) r.size(); i++)
        res[i + k] += r[i];
    for (int i = 0; i < (int) a1b1.size(); i++)
        res[i] += a1b1[i];
    for (int i = 0; i < (int) a2b2.size(); i++)
        res[i + n] += a2b2[i];
    return res;
}

bigint operator*(const bigint &v) const {
    vector<int> a6 = convert_base(this->a, base_digits, 6);
    vector<int> b6 = convert_base(v.a, base_digits, 6);
    vll a(a6.begin(), a6.end());
    vll b(b6.begin(), b6.end());
    while (a.size() < b.size())
        a.push_back(0);
```

```
while (b.size() < a.size())
    b.push_back(0);
while (a.size() & (a.size() - 1))
    a.push_back(0), b.push_back(0);
vll c = karatsubaMultiply(a, b);
bigint res;
res.sign = sign * v.sign;
for (int i = 0, carry = 0; i < (int) c.size(); i++) {
    long long cur = c[i] + carry;
    res.a.push_back((int) (cur % 1000000));
    carry = (int) (cur / 1000000);
}
res.a = convert_base(res.a, 6, base_digits);
res.trim();
return res;
}
};
```

Struktury danych (4)

min-que.cpp
Opis: Zamortyzowane $O(n)$. Dodawanie do końca, usuwanie z przodu i podaj min.

```
struct minQue
{
    deque<pair<int, int> > q;
    int _size = 0;
    void push(int val)
    {
        _size++;
        pair<int, int> nq = {val, 1};
        while(!q.empty() && q.back().first >= val) //zmienic na <=
            jezeeli ma byc maxQue
        {
            nq.second += q.back().second;
            q.pop_back();
        }
        q.push_back(nq);
    }
    void pop()
    {
        _size--;
        assert(!q.empty());
        if(q.front().second > 1) q.front().second--;
        else q.pop_front();
    }
    int top()
    {
        return q.front().first;
    }
    int size()
    {
        return _size;
    }
};
```

treap.cpp
Opis: $O(\log n)$ czasowo. Treap z operacjami na (multi)secie. kth(t, k) zwraca k-tą najmniejszą liczbę.

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int val, rank;
        int size;
```

```
Node(): left(nullptr), right(nullptr), val(0), rank((int)rng
    ()), size(0) {}
};
using pNode = Node*;

Node pool[1'000'003];
int curr = 0;

int sz(pNode t) {
    return t == nullptr ? 0 : t->size;
}

void upd(pNode t) {
    t->size = sz(t->left) + sz(t->right) + 1;
}

pNode merge(pNode a, pNode b) {
    if(a == nullptr) return b;
    if(b == nullptr) return a;

    if(a->rank < b->rank) {
        a->right = merge(a->right, b);
        upd(a);
        return a;
    }
    else {
        b->left = merge(a, b->left);
        upd(b);
        return b;
    }
}

pair<pNode, pNode> split(pNode t, int x) {
    if(t == nullptr) return {nullptr, nullptr};

    if(t->val <= x) {
        auto [a, b] = split(t->right, x);
        t->right = a;
        upd(t);
        return {t, b};
    }
    else {
        auto [a, b] = split(t->left, x);
        t->left = b;
        upd(t);
        return {a, t};
    }
}

void insert(pNode &root, int x) {
    pNode u = pool + (curr++);
    u->val = x;
    u->size = 1;
    auto [a, b] = split(root, x);
    root = merge(merge(a, u), b);
}

bool find(pNode t, int x) {
    if(t == nullptr) return false;
    if(x == t->val) return true;
    if(x < t->val) return find(t->left, x);
    else return find(t->right, x);
}

void erase(pNode root, int x) {
    auto [ab, c] = split(root, x);
    auto [a, b] = split(ab, x-1);
    root = merge(a, c);
}
```

```
int kth(pNode t, int k) {
    assert(k >= 0);
    if(sz(t->left) == k-1) return t->val;
    if(k > sz(t->left)) return kth(t->right, k-1-sz(t->left));
    else return kth(t->left, k);
}
}
```

implicit-treap.cpp

Opis: $O(\log n)$ czasowo. Implicit treap z podstawowymi operacjami, przykładowo odwracanie na przedziale (może być dowolna operacja). insert(root, k, x) wstawia element o wartości x w k -te miejsce tablicy. rev(root, l, r) odwraca przedział $[l; r]$ (indeksy od 0).

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int rank;
        int size;
        bool rev;
        int val;
        Node(): left(nullptr), right(nullptr), rank(int(rng())),
            size(0), rev(false), val(0) {}
    };
    using pNode = Node*;

    Node pool[300'003];
    int curr = 0;

    int sz(pNode t) {
        return t == nullptr ? 0 : t->size;
    }

    void upd(pNode t) {
        t->size = sz(t->left) + sz(t->right) + 1;
    }

    void push_lazy(pNode t) {
        if(t->rev == false) return;
        if(t->left) t->left->rev = !t->left->rev;
        if(t->right) t->right->rev = !t->right->rev;
        swap(t->left, t->right);
        t->rev = false;
    }

    pNode merge(pNode a, pNode b) {
        if(a == nullptr) return b;
        if(b == nullptr) return a;

        push_lazy(a);
        push_lazy(b);
        if(a->rank < b->rank) {
            a->right = merge(a->right, b);
            upd(a);
            return a;
        }
        else {
            b->left = merge(a, b->left);
            upd(b);
            return b;
        }
    }

    pair<pNode, pNode> split(pNode t, int x) {
        if(t == nullptr) return {nullptr, nullptr};

        push_lazy(t);
```

```
        if(sz(t->left)+1 <= x) {
            auto [a, b] = split(t->right, x-(sz(t->left)+1));
            t->right = a;
            upd(t);
            return {t, b};
        }
        else {
            auto [a, b] = split(t->left, x);
            t->left = b;
            upd(t);
            return {a, t};
        }
    }

    pNode build(vector<int> a) {
        pNode root = nullptr;
        int n = ssize(a);
        for(int i = 0; i < n; ++i) {
            pNode u = pool + i;
            u->size = 1;
            u->val = a[i];
            root = merge(root, u);
        }
        curr += n;
        return root;
    }

    void insert(pNode &root, int k, int x) {
        auto [a, b] = split(root, k-1);
        Node *u = pool + (curr++);
        u->size = 1;
        u->val = x;
        root = merge(merge(a, u), b);
    }

    void reverse(pNode &root, int l, int r) {
        auto [ab, c] = split(root, r+1);
        auto [a, b] = split(ab, l);
        b->rev = !b->rev;
        root = merge(a, merge(b, c));
    }

    void print(pNode t) {
        if(t == nullptr) return;
        push_lazy(t);
        print(t->left);
        cout << t->val << ' ';
        print(t->right);
    }
}
```

uf-rollback.cpp

Opis: $O(\log n)$ czasowo. Union find z roll backami – roll_back(x) cofa ostatnie x operacji unionn (union wierzchołków z tego samego zbioru się nie liczy).

```
struct dynamicUF {
    vector<int> e;
    dynamicUF(int n):e(n, -1) {}
    stack<array<int, 3>> s;

    int get(int a) {
        while(e[a] >= 0) a = e[a];
        return a;
    }

    int unionn(int a, int b) {
        a = get(a); b = get(b);
        if(a == b) return 0;
```

```
        if(-e[a] > -e[b]) swap(a, b);
        s.push({a, b, e[a]});
        e[b] += e[a];
        e[a] = b;
        return 1;
    }

    void roll_back(int m = 1) {
        while (m--) {
            auto [a, b, sz] = s.top();
            s.pop();
            e[b] -= sz;
            e[a] = sz;
        }
    }
};
```

fenwick-tree.cpp

Opis: (Binary Indexed Tree) $O(\log N)$, indeksowane od 0. Update w punkcie, zapytanie na przedziale

```
struct BIT {
    vector<int> tree;
    int N;
    BIT() {N = 0;}
    BIT(int n) {
        N = n;
        tree.resize(N);
    }
    void update(int p, int x) {
        for (++p; p <= N; p += (p&-p)) tree[p-1] += x;
    }
    int sum(int r) {
        int res = 0;
        for (; r > 0; r -= (r&-r)) res += tree[r-1];
        return res;
    }
    int query(int l, int r) {return sum(r+1)-sum(l);}
    int lower_bound(int sum) {
        int pos = 0;
        for (int pw = 1<<25; pw; pw>>=1) {
            int npos = pos+pw;
            if (npos <= N && tree[npos-1] < sum) {
                pos = npos; sum -= tree[pos-1];
            }
        }
        return pos;
    }
};
```

range.cpp

Opis: $O(\log n)$. Dodaj na przedziale, podaj wartość w punkcie.

```
#include "fenwick-tree.cpp" //keep-include

struct RangeAdd {
    BIT diff;
    RangeAdd(int n): diff(n) {}
    void update(int l, int r, int x) {
        if (l) diff.update(l-1, x);
        diff.update(r, -x);
    }

    int query(int p) {
        return diff.sum(p);
    }
};
```

fenwick-tree2d.cpp

Opis: $O(\log^2 n)$. Dodaj w punkcie, suma na prostokącie.

```
struct BIT2d {
    vector<vector<int>> tree;
    int n, m;
    BIT2d() {n = m = 0;}
    BIT2d(int N, int M) {
        n = N; m = M;
        tree.resize(n, vector<int>(m));
    }
    void update(int x, int y, int v) {
        for (++x; x <= n; x += (x&-x))
            for (int p = y+1; p <= m; p += (p&-p))
                tree[x-1][p-1] += v;
    }
    int sum(int x, int y) {
        int res = 0;
        for (; x; x -= (x&-x))
            for (int j = y; j; j -= (j&-j))
                res += tree[x-1][j-1];
        return res;
    }
    int query(int x1, int y1, int x2, int y2) {return sum(x2+1,
        y2+1)-sum(x1, y2+1)-sum(x2+1, y1)+sum(x1, y1);}
};
```

pbds.cpp

Opis: Ordered set.

```
#include <ext/pb_ds/assoc_container.hpp> //keep-include
#include <ext/pb_ds/tree_policy.hpp> //keep-include
using namespace __gnu_pbds;

template<typename T> using ordered_set = tree<
T,
null_type,
less<T>,
rb_tree_tag,
tree_order_statistics_node_update>;

void test() {
    ordered_set<int> S;
    S.insert(1);
    cout << *S.find_by_order(0) << '\n'; //zwraca k-ty element (
        indeksowane od 0)
    cout << (S.end() == find_by_order(1)) << '\n';
    cout << S.order_of_key(4) << '\n'; //zwraca liczbe elementow
        mniejszych od x
}
```

merge-sort-tree.cpp

Opis: Preprocessing $O(n)$, query w $O(\log n \cdot T(n))$, gdzie $T(n)$ to czas op-
eracji na przedziale bazowym. Pamięć $O(n \log n)$.

```
const int base = 1<<20;
struct mergeSortTree
{
    vector<vector<int> > tree;
    void build(int w, vector<int>& A)
    {
        if(w >= base)
        {
            if(w - base < A.size()) tree[w].push_back({A[w-base]});
            return;
        }
        build(w*2, A);
        build(w*2+1, A);
        merge(tree[w*2].begin(), tree[w*2].end(),
```

```
        tree[w*2+1].begin(), tree[w*2+1].end(),
        back_inserter(tree[w]));
        return;
    }
    mergeSortTree(vector<int>& A)
    {
        tree.resize(base*2);
        build(1, A);
    }
    int query(int w, int p, int k, int x, int y, int val)
    {
        if(p > y || k < x) return 0; //poza zapytaniem
        if(x <= p && k <= y)
        {
            auto it = upper_bound(tree[w].begin(), tree[w].end(), val
                );
            //tutaj robimy cos na przedziale bazowym zapytania
            return distance(tree[w].begin(), it); //returnujemy wynik
                tego przedzialu
        }
        return query(w*2, p, (p+k)/2, x, y, val) +
            query(w*2+1, (p+k)/2+1, k, x, y, val); //tu dowolna
            operacja (max, +, -, *, czy cokolwiek)
    }
    int query(int x, int y, int val) {return query(1, 0, base-1,
        x, y, val);} //ile mniejszych od val na (x, y)
};
```

online-cht.cpp

Opis: Dodawanie prostej i query w $O(\log n)$. Daje min z funkcji liniowych.

```
struct Line {
    bool type;
    double x;
    ll m, c;
};

bool operator<(Line l1, Line l2) {
    if (l1.type || l2.type) return l1.x < l2.x;
    return l1.m > l2.m;
}

struct CHT{
    set<Line> cht;
    bool has_prev(set<Line>::iterator it) { return it != cht.
        begin(); }
    bool has_next(set<Line>::iterator it) {
        return it != cht.end() && next(it) != cht.end();
    }
    double intersect(set<Line>::iterator l1, set<Line>::iterator
        l2) {
        return (double) (l1->c - l2->c) / (l2->m - l1->m);
    }
    void calc_x(set<Line>::iterator it) {
        if (has_prev(it)) {
            Line l = *it;
            l.x = intersect(prev(it), it);
            cht.insert (cht.erase(it), l);
        }
    }
    bool bad(set<Line>::iterator it) {
        if (has_next(it) && next(it)->c <= it->c) return true;
        return (has_prev(it) && has_next(it) && intersect(prev(it),
            next(it)) <= intersect(prev(it), it));
    }
    void add_line(ll m, ll c) {
        set<Line>::iterator it;
        it = cht.lower_bound({0, 0, m, c});
        if (it != cht.end() && it->m == m) {
```

```
            if (it->c <= c) return;
            cht.erase(it);
        }
        it = cht.insert({0, 0, m, c}).first;
        if (bad(it)) cht.erase(it);
        else {
            while (has_prev(it) && bad(prev(it))) cht.erase(prev(it))
                ;
            while (has_next(it) && bad(next(it))) cht.erase(next(it))
                ;
            if (has_next(it)) calc_x(next(it));
            calc_x(it);
        }
    }
    ll query(ll h) {
        Line l = *prev(cht.upper_bound({1, (double)h, 0, 0}));
        return l.m * h + l.c;
    }
};
```

wavelet.cpp

Opis: $O(\log^2 \text{ALPH})$ czasowo, $O(n \log \text{ALPH})$ pamięciowo. Constructor
przyjmuje ciąg na którym rozpinamy drzewo, rank(l, r, k) zwraca k-tą
najmniejszą liczbę na przedziale [l; r]. (Uwaga!) pamięć około 262 Mb, dla
 $N, Q \leq 2 \cdot 10^5$.

```
//funkcja cnt jest w  $O(\log)$  co daje cale operacje w  $O(\log^2)$ ,
da sie cnt w czasie stalym — do dopisania
struct wavelet_tree {
    const static int ALPH = (1<<18)-1;
    struct Node {
        int low, high, mid;
        vector<int> v;
        Node *left, *right;

        Node() {
            low = 0, high = ALPH;
            left = right = nullptr;
            mid = (low+high)/2;
        }

        Node (const vector<int> &x, const vector<int> &t, int a =
            0, int b = ALPH): low(a), high(b), v(x) {
            mid = ((ll)low+high)/2;
            if(ssize(x) && low != high) {
                vector<int> div[2];
                for(auto e : v) div[t[e]<=mid].eb(e);
                if(ssize(div[1])) left = new Node(div[1], t, low, mid);
                else left = nullptr;
                if(ssize(div[0])) right = new Node(div[0], t, mid+1,
                    high);
                else right = nullptr;
            }
            else {
                left = right = nullptr;
            }
        }
    };

    Node *root;
    wavelet_tree(const vector<int> &b) {
        vector<int> idx;
        REP(i, ssize(b)) idx.eb(i);
        root = new Node(idx, b);
    };

    int cnt(Node *v, int l, int r) {
        if(v == nullptr) return 0;
```

```
        return int (upper_bound(all (v->v), r)-lower_bound(all (v->v),
1));
    }

    int rank(int l, int r, int k, Node *v) {
        if (v->low == v->high) return v->low;
        if (cnt(v->left, l, r) >= k) return rank(l, r, k, v->left);
        else return rank(l, r, k-cnt(v->left, l, r), v->right);
    }
    int rank(int l, int r, int k) {
        return rank(l, r, k, root);
    }

    void clear(Node *v) {
        if (v->left != nullptr) clear(v->left);
        if (v->right != nullptr) clear(v->right);
        delete v;
    }
    ~wavelet_tree() {
        clear(root);
    }
};
```

Grafy (5)

5.1 Drzewo DFS

Mamy drzewo dfs z obliczonymi czasami wejścia w tablicy **pre** i funkcją

$$\text{low}[v] = \min \begin{cases} \text{pre}[v], \\ \text{pre}[u] & \text{dla krawędzi niedrzewowych } (u, v), \\ \text{low}[u] & \text{dla krawędzi drzewowych } (u, v). \end{cases}$$

Wierzchołek *v* jest punktem artykulacji jeżeli

$$\text{low}[u] \geq \text{pre}[v]$$

dla dowolnego *u* będącego synem *v* w drzewie dfs.

Krawędź (*v*, *u*) (*v* jest ojcem *u*) jest mostem, jeżeli jest krawędzią drzewową i

$$\text{low}[u] > \text{pre}[v].$$

hld.cpp

Opis: *O*(*n*) preprocessing, update i query w *O*(log² *n*). Update to zmiana wartosci w wierzcholku, query to max na sciezce. Drzewo przedzialowe do dopisania tak jak w komentarzu.

```
const int inf = 1e9+10;

//struct segtree {
//    segtree(int n = 0) {}
//    void update(int v, int x) {}
//    int query(int l, int r) {}
//};

struct HLD {
    int n;
    vector<vector<int>>> g;
    segtree t;
    int timer;
```

```
vector<int> sz, d, parent, path, c, pre;

HLD(const vector<vector<int>>> &graph): n(ssize(graph)), g(
graph) {
    sz = d = parent = path = c = pre = vector<int> (n);

    d[0] = 0;
    dfs(0);

    path[0] = 0;
    timer = 0;
    make_hld(0);
    t = segtree(timer);
}

void dfs(int v, int p = -1) {
    parent[v] = p;
    sz[v] = 1;
    c[v] = -1;
    for (auto u : g[v]) {
        if (u == p) continue;
        d[u] = d[v] + 1;
        dfs(u, v);
        sz[v] += sz[u];
        if (c[v] == -1 || sz[c[v]] < sz[u]) c[v] = u;
    }
}

void make_hld(int v, int p = -1) {
    pre[v] = timer++;
    if (c[v] != -1) {
        path[c[v]] = path[v];
        make_hld(c[v], v);
    }
    for (auto u : g[v]) {
        if (u == p || u == c[v]) continue;
        path[u] = u;
        make_hld(u, v);
    }
}

void update(int v, int x) {
    t.update(pre[v], x);
}

int query(int a, int b) {
    int res = -inf;
    while (path[a] != path[b]) {
        if (d[path[a]] < d[path[b]]) swap(a, b);
        res = max(res, t.query(pre[path[a]], pre[a]));
        a = parent[path[a]];
    }
    if (d[a] < d[b]) swap(a, b);
    res = max(res, t.query(pre[b], pre[a]));
    return res;
}
};
```

scc.cpp

Opis: silnie spójne skladowe. Argument do constructora to graf skierowany w postaci vector<vector<int>>> v. Oblicza vector indeksów spójnie skladowej do której należy *v* o nazwie scc. Czas, pamiec *O*(*n* + *m*).

```
struct SCC{
    vector<vector<int>>>v,x;
    int n;
    queue<int>q;
    vector<bool>vis;
```

```
int kt=0;
vector<int>scc;
SCC(const vector<vector<int>>>&vv):v(vv),n(ssize(vv)){
    vis.resize(n,0);
    scc.resize(n);
    for(int i=0;i<n;i++){
        if(!vis[i])dfs0(i);
    }
    make_x();
    fill(all(vis),0);
    while(!q.empty()){
        if(!vis[q.front()]){
            dfs1(q.front());
            kt++;
        }
        q.pop();
    }
}
void dfs0(int a){
    vis[a]=1;
    for(auto b:v[a]){
        if(!vis[b])dfs0(b);
    }
    q.push(a);
}
void make_x(){
    x.resize(n);
    for(int i=0;i<ssize(v);i++){
        for(auto b:v[i]){
            x[b].push_back(i);
        }
    }
}
void dfs1(int a){
    vis[a]=1;
    scc[a]=kt;
    for(auto b:v[a]){
        if(!vis[b])dfs1(b);
    }
}
};
```

bellman-ford.cpp

Opis: *W* *O*(*nm*) oblicza tablice **D** z odległościami od *s* w grafie skierowanym ważonym (dopuszczane ujemne wagi). Po obliczeniu odległości isNegCyc() zwraca czy w grafie jest ujemny cykl.

```
const int INF = numeric_limits<int>::max();
struct edge
{
    int a, b, cost;
    edge(int _a, int _b, int _cost){a = _a; b = _b; cost = _cost;
}
};
struct Bellman_Ford
{
    int n;
    vector<edge> edges;
    vector<int> D;
    bool makeStep()
    {
        bool e = 0;
        for(auto v : edges)
            if(D[v.a] != INF)
                if(D[v.b] > D[v.a] + v.cost)
                    D[v.b] = D[v.a] + v.cost, e = true;
        return e;
    }
    Bellman_Ford(int _n, vector<edge>& _edges, int s)
```

```
{
    n = _n; edges = _edges;
    D.resize(n+1, INF);
    D[s] = 0;
    while(_n-->1) makeStep();
}
bool isNegCyc()
{
    return makeStep();
}
};
```

5.2 Floyd-Warshall

$O(n^3)$. Oblicza odległość między każdą parą wierzchołków w skierowanym grafie ważonym (dopuszczalne wagi ujemne).

```
Dla k = 0 do n - 1
    Dla i = 0 do n - 1
        Dla j = 0 do n - 1
            Odleglosc[i, j] = min(Odleglosc[i, j],
Odleglosc[i, k] + Odleglosc[k, j])
```

Tekstówki (6)

kmp.cpp

Opis: $O(n + m)$. Constructor przyjmuje wzorzec i tekst, oblicza vector indeksów wystąpień wzorca w tekście.

```
struct KMP{
    string s;
    vector<int>pi;
    vector<int>ans;
    KMP(string &a,string &b):s(a+'#'+b){
        prefsuf();
        for(int i=ssize(a);i<ssize(s);i++){
            if(pi[i]==ssize(a)) ans.push_back(i-2*ssize(a));
        }
    }
    void prefsuf(){
        int n=ssize(s);
        pi.resize(n,0);
        for(int i=1;i<n;i++){
            int j=pi[i-1];
            while(j>0&&s[i]!=s[j]) j=pi[j-1];
            pi[i]=j+(s[i]==s[j]);
        }
    }
};
```

z.cpp

Opis: $O(n)$. Zwraca tablice z, gdzie $z[i]$ jest równe najdłuższemu wspólnemu prefiksowi słowa s i sufiksu $[i, \dots, n - 1]$.

```
vector<int> z(string s) {
    int n = ssize(s);
    vector<int> z(n);

    int l = 0, r = 0;
    for (int i = 1; i < n; ++i) {
        if (i < r) z[i] = min(z[i-1], r-i);
        while (i + z[i] < n && s[i+z[i]] == s[z[i]]) ++z[i];
        if (i + z[i] > r) {
            r = i + z[i];
            l = i;
        }
    }
}
```

```
return z;
}
```

suffix-array.cpp

Opis: (źródło: <https://github.com/tonowak/acmlib/blob/master/code/string/suffix-array-short/main.cpp>) $O(n \log n)$, zawiera posortowane suffixy, zawiera pusty suffix, $lcp[i]$ to lcp suffixu $sa[i - 1]$ i $sa[i]$. Dla $s = aabaaab$, $sa=\{7, 3, 4, 0, 5, 1, 6, 2\}$, $lcp=\{0, 0, 2, 3, 1, 2, 0, 1\}$

```
pair<vector<int>, vector<int>> suffix_array(vector<int> s, int
alpha = 26) {
    ++alpha;
    for(int &c : s) ++c;
    s.emplace_back(0);
    int n = ssize(s), k = 0, a, b;
    vector<int> x(s.begin(), s.end());
    vector<int> y(n), ws(max(n, alpha)), rank(n);
    vector<int> sa = y, lcp = y;
    iota(sa.begin(), sa.end(), 0);
    for(int j = 0, p = 0; p < n; j = max(1, j * 2), alpha = p) {
        p = j;
        iota(y.begin(), y.end(), n - j);
        for (int i = 0; i < n; ++i) if(sa[i] >= j)
            y[p++] = sa[i] - j;
        fill(ws.begin(), ws.end(), 0);
        for (int i = 0; i < n; ++i) ws[x[i]]++;
        for (int i = 1; i <= alpha - 1; ++i) ws[i] += ws[i - 1];
        for(int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
        swap(x, y);
        p = 1, x[sa[0]] = 0;
        for (int i = 1; i <= n-1; ++i) a = sa[i - 1], b = sa[i], x[
            b] =
                (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
    }
    for(int i = 1; i <= n-1; ++i) rank[sa[i]] = i;
    for(int i = 0, j; i < n - 1; lcp[rank[i+]] = k)
        for(k && k--, j = sa[rank[i] - 1];
            s[i + k] == s[j + k]; k++);
    lcp.erase(lcp.begin());
    return {sa, lcp};
}
```

Flow (7)

dinic.cpp

Opis: $O(V^2E)$, chyba że mamy sieć jednostkową (max flow na krawędzi to 1) to $O(E\sqrt{V})$.

```
const int INF = numeric_limits<int>::max();
struct flowEdge
{
    int u, v, cap, flow;
    flowEdge(int _u, int _v, int _cap, int _flow){u = _u; v = _v;
        cap = _cap; flow = _flow;}
    flowEdge(){}
};
struct dinic
{
    vector<flowEdge> edges;
    vector<vector<int>> > idx;
    queue<int> q;
    vector<int> lvl;
    vector<int> ptr;
    int n, m;
    dinic(int _n, int _m, vector<pair<pair<int, int>, int> >&
        _edges) //krawedzie musza byc znormalizowane do DG
    {
        n = _n; m = _m;
```

```
idx.resize(n+1);
lvl.resize(n+1);
ptr.resize(n+1);
for(auto v : _edges)
    edges.push_back(flowEdge(v.first.first, v.first.second, v
        .second, 0));
    edges.push_back(flowEdge(v.first.second, v.first.first,
        0, 0));
    idx[v.first.first].push_back(edges.size()-2);
    idx[v.first.second].push_back(edges.size()-1);
}
}
bool bfs()
{
    fill(lvl.begin(), lvl.end(), -1);
    q.push(0);
    lvl[0] = 0;
    while(!q.empty())
    {
        int a = q.front();
        q.pop();
        for(auto v : idx[a])
        {
            if(lvl[edges[v].v] != -1) continue;
            if(edges[v].cap - edges[v].flow < 1) continue;
            lvl[edges[v].v] = lvl[a]+1;
            q.push(edges[v].v);
        }
    }
    return lvl[n] != -1;
}
int dfs(int v, int aktmin)
{
    if(v == n) return aktmin;
    if(aktmin == 0) return aktmin;
    for(int& i = ptr[v];i<idx[v].size();i++)
    {
        int id = idx[v][i];
        int nv = edges[id].v;
        if(lvl[v] + 1 != lvl[nv] || edges[id].cap - edges[id].
            flow < 1) continue;
        long long bFlow = dfs(nv, min(aktmin, edges[id].cap -
            edges[id].flow));
        if(bFlow == 0) continue;
        edges[id].flow += bFlow;
        edges[id^1].flow -= bFlow;
        return bFlow;
    }
    return 0;
}
int maxFlow()
{
    int res = 0;
    while(true)
    {
        if(!bfs()) break;
        fill(ptr.begin(), ptr.end(), 0);
        while(int flow = dfs(0, INF)) res += flow;
    }
    return res;
}
};
```

Optymalizacje (8)

fastio.cpp

Opis: fastio.

```
inline void SuckInt(int *n){
    register char c=0;
    register int r=1;
    (*n)=0;
    while (c<33) c=getc_unlocked(stdin);
    if (c==45)
        r=-1, c=getc_unlocked(stdin);
    while (c>32)
        (*n)=(*n)*10+c-48, c=getc_unlocked(stdin);
    (*n)*=r;
}

inline void SuckLong(long long *n){
    register char c=0;
    register int r=1;
    (*n)=0;
    while (c<33) c=getc_unlocked(stdin);
    if (c==45)
        r=-1, c=getc_unlocked(stdin);
    while (c>32)
        (*n)=(*n)*10+c-48, c=getc_unlocked(stdin);
    (*n)*=r;
}

inline void ThrowSlow(string s){
    for (auto i: s)
        putc_unlocked(i, stdout);
}

inline void Throw(int n, bool f=true){
    if (n<0 && f) putc_unlocked('-', stdout), n*=-1;
    if (n==0 && f) putc_unlocked('0', stdout);
    if (n==0) return;
    Throw(n/10, 0);
    putc_unlocked(n%10+'0', stdout);
}
```

allocator.cpp

Opis: Potencjalnie szybszy alokator, jezeli dopychamy kolanem, szczegolnie dziala gdy alokujemy duzo malych vectorow.

```
namespace
{
    char buf[400'000'000];
    char *buf_ptr = buf;
}

template<typename T> struct my_allocator {
    using value_type = T;
    using size_type = size_t;

    T *allocate (size_type count) {
        if(count * sizeof(T) <= 4096) {
            T *result = (T *) buf_ptr;
            buf_ptr += count * sizeof(T);
            return result;
        } else {
            return new T[count];
        }
    }

    void deallocate(T *pointer, size_type count) {
        //delete[] pointer;
    }
};

vector<int, my_allocator<int>> v; //zamiast vector<int> v;
```

pragmy.cpp

Opis: pragmy.

```
#pragma GCC optimize("Ofast")
#pragma GCC target("avx2")
```